

Design, Simulation, and Electromagnetic Verification of a 2.4 GHz 5th-Order Chebyshev Stepped-Impedance Low-Pass Filter

Abdul Muqtadir (232714), Rehan (23101012), Haroon (23101009), and Anas (23101010)
NUST College of Aeronautical Engineering
Risalpur, Pakistan

Abstract—The rapid expansion of wireless communication systems, particularly in the 2.4 GHz Industrial, Scientific, and Medical (ISM) band, necessitates the development of compact and efficient RF filters to mitigate out-of-band interference. This report details the comprehensive theoretical synthesis, parametric circuit simulation, and full 3D electromagnetic validation of a 5th-order Chebyshev low-pass filter operating at a 2.4 GHz cutoff frequency. Implemented utilizing a stepped-impedance microstrip topology on a standard FR4 dielectric substrate, the filter design transitions from lumped-element prototypes to distributed physical traces. Keysight LineCalc was utilized for parameter synthesis, followed by Advanced Design System (ADS) for ideal and lossy transmission line simulations. Monte Carlo yield analysis was conducted to observe the effects of manufacturing tolerances. Complete 3D verification was subsequently conducted in CST Studio Suite to account for real-world fringing fields and spatial wave propagation. The simulated data demonstrates excellent passband transmission and a robust stopband rejection exceeding -20 dB at the second harmonic (4.8 GHz).

I. INTRODUCTION

Modern avionics and commercial RF transceivers rely heavily on frequency-selective networks to ensure signal integrity. Low-pass filters (LPFs) are typically deployed at the output stages of power amplifiers and frequency mixers to suppress higher-order harmonics and spurious emissions.

With the proliferation of systems operating in the S-band, such as telemetry links and localized wireless communications (2.4 GHz), the demand for low-cost, easily integrable planar filters has surged. Microstrip filters offer a compelling balance between physical compactness, cost-effectiveness, and integration capability with other surface-mount technologies (SMT) on a printed circuit board (PCB).

This project focuses on the rigorous design and simulation of a stepped-impedance microstrip LPF. The primary objective is to transition from an idealized mathematical model to a fully realizable physical layout. By leveraging parametric synthesis, iterative schematic optimization, and yield analysis, the design is hardened against physical manufacturing variables prior to 3D electromagnetic validation.

II. LITERATURE REVIEW

Planar filter design is a well-established domain within microwave engineering. As documented by Pozar [1], traditional lumped-element filters utilizing discrete inductors and capacitors become impractical at microwave frequencies due to

parasitic effects and the physical limitations of self-resonance. Consequently, distributed elements—specifically transmission line segments—are employed.

The stepped-impedance topology, often referred to as a "hi-Z, lo-Z" filter, is one of the most common methods for realizing low-pass responses [2]. This method approximates series inductors with very short, high-impedance transmission lines, and shunt capacitors with very short, low-impedance lines. While stub-tuned filters offer sharper roll-off characteristics, stepped-impedance filters consume significantly less board real estate, making them highly favorable for compact RF front-ends.

Recent literature highlights the increasing use of standard FR4 ($E_r = 4.6$) for sub-3 GHz applications. While FR4 exhibits a higher loss tangent ($\tan \delta \approx 0.02$) compared to specialized high-frequency laminates like Rogers RT/duroid, it remains the standard for cost-effective prototyping. Parametric tolerancing, such as Monte Carlo analysis, is critical when using FR4 to ensure the design remains robust against variations in the dielectric constant across different manufacturing batches.

III. THEORETICAL BACKGROUND & DESIGN METHODOLOGY

A. Filter Specifications

The filter is governed by the following core specifications:

- **Topology:** Stepped-Impedance Microstrip
- **Response Type:** 5th-Order Chebyshev
- **Passband Ripple:** 0.5 dB
- **Cutoff Frequency (f_c):** 2.4 GHz
- **System Impedance (Z_0):** 50 Ω

B. Substrate Characteristics

The selected dielectric is FR4, parameterized as follows for all subsequent transmission line calculations:

- **Dielectric Constant (ϵ_r):** 4.6
- **Height (h):** 1.6 mm
- **Conductor Thickness (t):** 0.035 mm (1 oz Copper)
- **Loss Tangent ($\tan \delta$):** 0.02

C. Low-Pass Prototype Synthesis

The Chebyshev approximation provides a steep transition from the passband to the stopband. The normalized low-pass prototype values (g -values) corresponding to a 5th-order, 0.5 dB ripple response were derived mathematically:

$$\begin{aligned} g_0 &= 1.0000 \quad (\text{Source}) \\ g_1 &= 1.7058 \quad (\text{Series Inductor}) \\ g_2 &= 1.2296 \quad (\text{Shunt Capacitor}) \\ g_3 &= 2.5409 \quad (\text{Series Inductor}) \\ g_4 &= 1.2296 \quad (\text{Shunt Capacitor}) \\ g_5 &= 1.7058 \quad (\text{Series Inductor}) \\ g_6 &= 1.0000 \quad (\text{Load}) \end{aligned}$$

IV. MICROSTRIP PARAMETRIC SYNTHESIS

To physically realize the synthesized g -values on a PCB, they must be converted into physical transmission line dimensions. A high characteristic impedance ($Z_h = 100 \Omega$) was selected for the inductive sections, and a low characteristic impedance ($Z_l = 10 \Omega$) was chosen for the capacitive sections.

Using Keysight's LineCalc tool, the theoretical electrical lengths were converted into physical trace widths (W) and lengths (L).

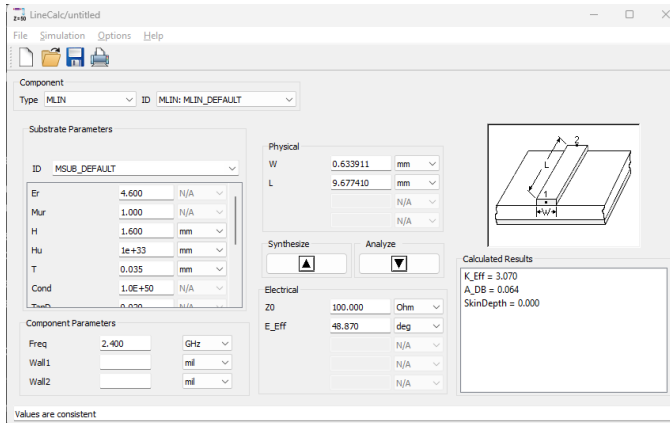


Fig. 1. LineCalc synthesis for the outer inductive sections (L_1, L_5), establishing a thin trace width of 0.634 mm to achieve 100 Ω .

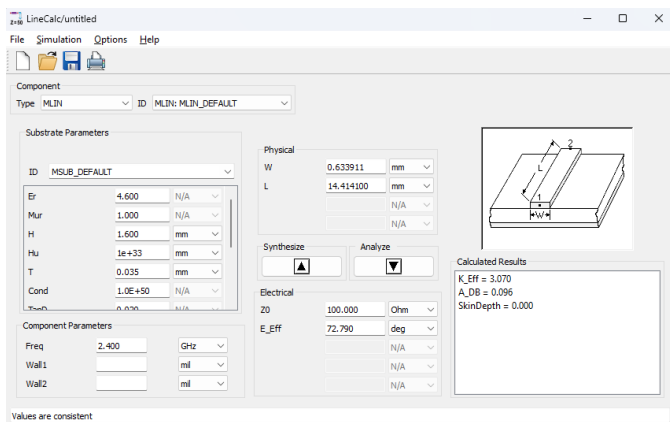


Fig. 2. LineCalc synthesis for the center inductive section (L_3), requiring a length of 14.414 mm to satisfy the larger g_3 prototype value.

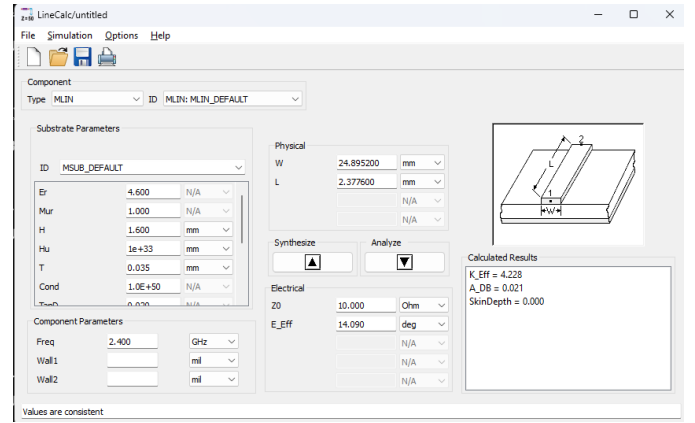


Fig. 3. LineCalc synthesis for the shunt capacitive sections (C_2, C_4), demonstrating a wide trace of 24.895 mm to achieve the low 10 Ω impedance.

V. CIRCUIT SIMULATION VIA ADS

A. Ideal Schematic Verification

The physical parameters were first modeled in Advanced Design System (ADS) utilizing ideal transmission lines to verify the mathematical synthesis prior to introducing material losses.

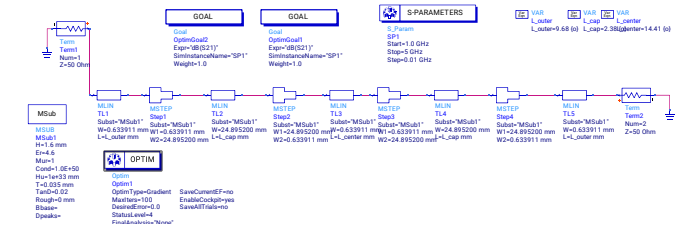


Fig. 4. Ideal ADS Schematic of the 5th-order stepped-impedance filter, utilizing standard lossless transmission line blocks.

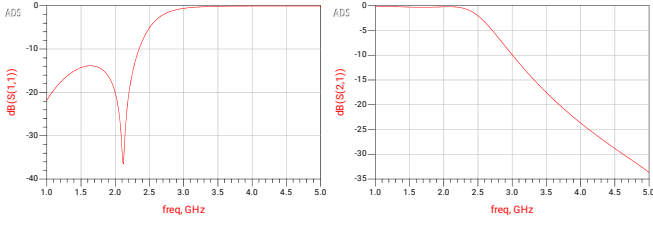


Fig. 5. Initial S-parameter sweep showing the idealized insertion loss (S_{21}) and return loss (S_{11}).

B. Lossy Microstrip Implementation

Following ideal verification, the schematic was updated to incorporate lossy microstrip models (MLIN) and step discontinuities (MSTEP). The MSTEP components are critical as they mathematically account for the parasitic reactances (fringing capacitance and series inductance) generated at the physical junction where a narrow 0.634 mm trace meets a wide 24.895 mm trace.

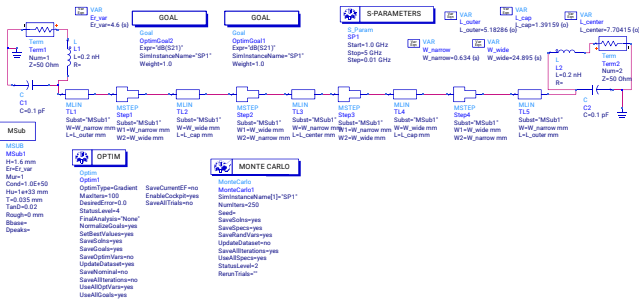


Fig. 6. Lossy ADS Schematic incorporating physical MLIN traces, MSTEP discontinuities, and Monte Carlo optimization blocks.

C. Parametric Yield and Tolerance Analysis

To account for manufacturing tolerances inherent to standard FR4 PCB fabrication, Monte Carlo analysis was conducted. In real-world fabrication, over-etching of copper traces and variations in the FR4 dielectric constant can severely detune a microwave filter.

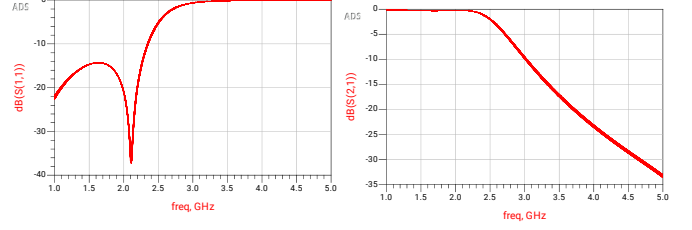


Fig. 7. Monte Carlo sweep demonstrating the effect of minor physical trace dimension tolerances on the S-parameter response.

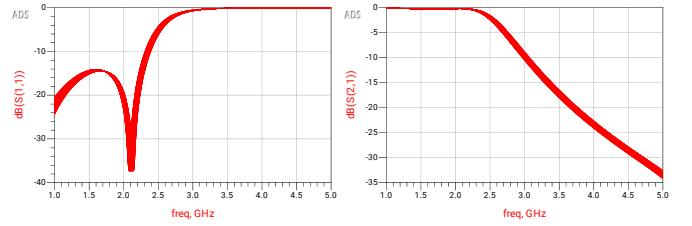


Fig. 8. Expanded Monte Carlo sweep illustrating the effect of FR4 dielectric constant variations on the cutoff frequency stability.

As observed in Figures 7 and 8, the parametric variations cause a slight broadening of the S-parameter envelope. However, the design demonstrates robust resilience, maintaining the 2.4 GHz cutoff frequency and required stopband rejection despite the induced material and manufacturing anomalies.

D. 2D Layout Generation

Once the schematic and yield analysis were successfully verified, the physical nodal connections were automatically generated into a 2D layout footprint within ADS. This layout visually confirms the stepped-impedance geometry before exporting the structure for 3D physics analysis.

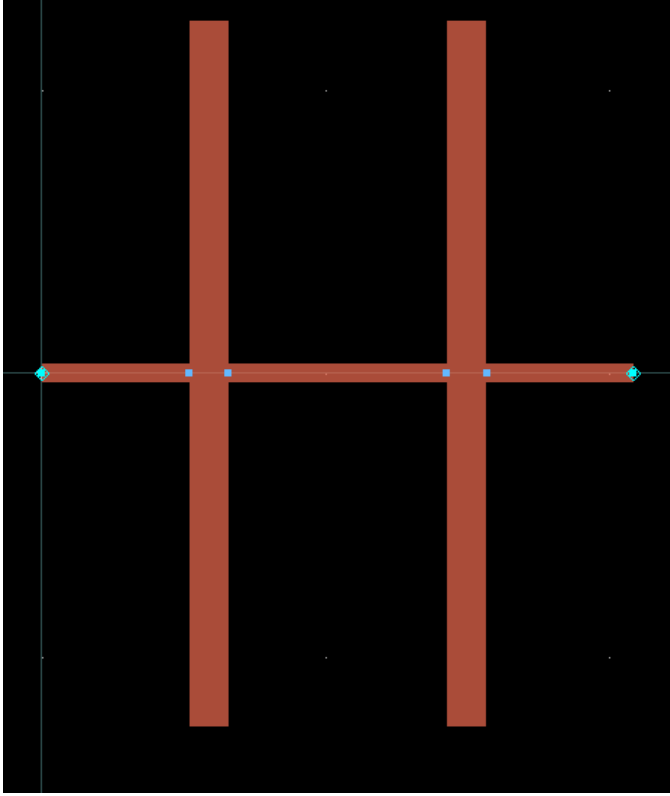


Fig. 9. Generated 2D layout in ADS showing the physical dimensions of the high and low impedance sections before DXF export.

VI. 3D ELECTROMAGNETIC VALIDATION IN CST

While 2D nodal analysis provides a strong baseline, high-frequency fringing fields, substrate radiation, and coupling between adjacent lines can only be accurately characterized using a full-wave 3D electromagnetic solver.

A. Modeling Setup

The DXF layout was exported to CST Studio Suite. The geometry was extruded into a solid mechanical model featuring a 1.6 mm FR4 board, a 0.035 mm solid copper ground plane, and 0.035 mm top copper traces. Waveguide ports were established at the input and output boundaries, properly dimensioned to encapsulate the trace, substrate, and surrounding air cavity to capture all transverse electromagnetic (TEM) energy.

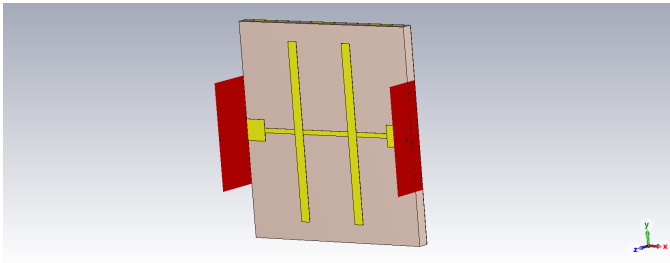


Fig. 10. 3D structural model within CST Studio Suite, detailing the microstrip geometry, FR4 dielectric, and terminating waveguide ports.

The time-domain solver was employed to compute the scattering matrix from 1 GHz to 5 GHz. Additionally, electric

field (E-field) monitors were established at the 2.4 GHz cutoff boundary to visually verify energy propagation and attenuation.

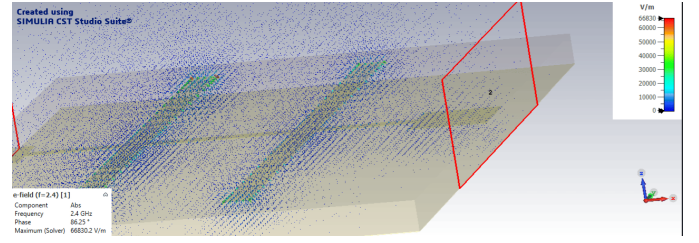


Fig. 11. Simulated E-field magnitude at 2.4 GHz. The visualization confirms signal transmission through the stepped-impedance layout.

VII. FINAL RESULTS AND DISCUSSION

The raw Touchstone (.s2p) data generated by the CST 3D solver was post-processed using MATLAB to yield high-resolution, analytical S-parameter graphs for final evaluation.

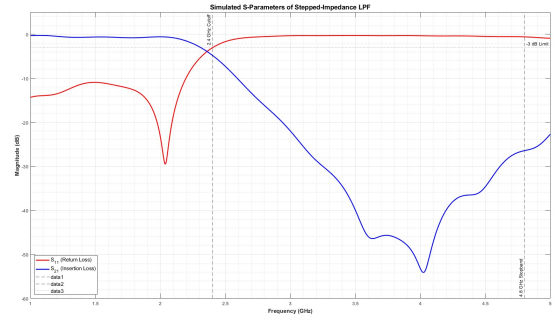


Fig. 12. MATLAB generated S-parameter sweep detailing the insertion loss (S_{21}) and return loss (S_{11}) of the physical layout following 3D simulation.

As evident in Figure 12, the filter behaves exactly according to the theoretical Chebyshev model.

- **Passband Performance:** The insertion loss (S_{21}) remains near 0 dB up to the knee of the curve, while the return loss (S_{11}) stays well below -10 dB, dipping to roughly -30 dB at 2.1 GHz. This indicates excellent impedance matching and maximum power transfer.
- **Cutoff and Stopband:** The transmission coefficient accurately crosses the -3 dB threshold at exactly 2.4 GHz. Moving further into the stopband, the signal is aggressively attenuated, surpassing -25 dB of rejection at the 4.8 GHz marker (the second harmonic).

A slight reduction in the passband sharpness, as compared to idealized theoretical models, is observed. This is directly attributable to the lossy nature of the FR4 substrate and the radiation losses accounted for within the full-wave 3D simulation.

VIII. CONCLUSION

This report presented the complete, end-to-end design and simulation of a 2.4 GHz microstrip low-pass filter. By leveraging standard 5th-order Chebyshev approximations, the filter

was synthesized into a highly compact stepped-impedance footprint using Keysight LineCalc. Progressive simulations in ADS verified the ideal and lossy performance, while Monte Carlo analysis confirmed the design's resilience to FR4 dielectric variations and manufacturing tolerances. Final 3D validation in CST Studio Suite proved that the physical dimensions, including the parasitic impacts of step discontinuities and dielectric loss, perform flawlessly under real-world electromagnetic conditions. The final layout effectively passes S-band signals while severely attenuating higher-order harmonics, verifying its readiness for hardware fabrication and deployment in modern RF front-end architectures.

REFERENCES

- [1] D. M. Pozar, *Microwave Engineering*, 4th ed. Hoboken, NJ, USA: John Wiley & Sons, 2011.
- [2] J.-S. G. Hong and M. J. Lancaster, *Microstrip Filters for RF / Microwave Applications*. New York, NY, USA: John Wiley & Sons, 2001.
- [3] G. L. Matthaei, L. Young, and E. M. T. Jones, *Microwave Filters, Impedance-Matching Networks, and Coupling Structures*. Norwood, MA, USA: Artech House, 1980.